

Design and Implementation of a Campus WiFi Sharing Platform Based on Evolutionary Game Theory

Yingrui Zhang, Haijiao Wang, Mingze Ren

Honors College, Northwestern Polytechnical University, Xi'an, China

Abstract—The uneven distribution of campus WiFi resources and the prevalence of free-riding behaviors frequently lead to a tragedy of the commons, severely impairing network utilization efficiency. To address this issue, this paper proposes and implements a campus WiFi sharing sandbox system based on Evolutionary Game Theory and mechanism design. Adopting a decoupled Browser / Server (B/S) architecture powered by a high-performance asynchronous backend, the system integrates a dual-asset Token and Reputation mechanism to precisely quantify user payoffs. An asymmetric dynamic reputation algorithm is designed to strictly penalize bandwidth consumption while heavily rewarding sharing, effectively establishing a cooperative surplus. Furthermore, a Generalized Second-Price (GSP) auction is deployed to guarantee fairness and truthful bidding during peak-hour bandwidth allocation. Most notably, a Multi-Agent System (MAS) is instantiated to visualize the macroscopic emergence of Cooperation, Free-riding, and Reciprocity strategies driven by Replicator Dynamics. Simulation results demonstrate that the proposed mechanisms successfully suppress free-riding anomalies, force credit-bankrupt agents to revert to cooperation, and enable the system to robustly converge to an Evolutionary Stable Strategy (ESS). This sandbox provides both theoretical validation and a pragmatic architectural reference for distributed resource sharing in campus networks.

Index Terms—Evolutionary Game Theory; Campus WiFi Sharing; Multi-Agent System; Generalized Second-Price Auction; Reputation Mechanism; Replicator Dynamics.

I. INTRODUCTION

WITH the deepening of campus informatization, WiFi has become an indispensable infrastructure for students' daily study and life. However, the uneven distribution of campus WiFi resources and the divergence in users' sharing willingness have severely impaired resource utilization efficiency. Evolutionary game theory has been widely adopted to address resource allocation challenges in wireless networks. In IRS-assisted communication, for example, Hu et al. [1] applied an evolutionary game approach to optimize reflection coefficients. Similarly, Yang et al. [2] integrated deep reinforcement learning with evolutionary game theory for dynamic spectrum access in cognitive radio networks. These works demonstrate that evolutionary game theory provides an effective framework for analyzing cooperative behaviors among distributed users.

The authors are with the Honors College, Northwestern Polytechnical University, Xi'an, China (email: zhangyingrui@mail.nwpu.edu.cn; wanghaijiao@mail.nwpu.edu.cn; renmingze@mail.nwpu.edu.cn)

The source code and experimental data are available at: [CampusWiFiGame](#)

From the perspective of the three core elements of game theory, the players are students, with three optional strategies: cooperation, free-riding, and reciprocity. User payoffs are quantified by the system's integral and credit mechanism: cooperation yields integral rewards and credit promotion; free-riding leads to integral deduction and credit reduction; reciprocity secures stable benefits through mutual cooperation.

To mitigate the above problems, this paper designs and implements a campus WiFi sharing game theory sandbox system, integrating game theory models with an integral credit system to visually demonstrate the evolution of cooperative, free-riding and reciprocal strategies. The system also employs a second-price auction mechanism for peak-hour resource allocation, inspired by auction-based user allocation approaches in heterogeneous networks [3]. This work provides theoretical verification and practical references for efficient campus WiFi resource sharing.

II. THEORETICAL FRAMEWORK

The system is grounded in three core theories: game theory, behavioral economics, and mechanism design. Evolutionary game theory serves as the primary foundation. Smith and Price [4] first established this framework, moving beyond the "complete rationality" assumption of classical game theory to better model strategic interactions among populations. It focuses on how group strategies evolve over time, enabling precise modeling of cooperative, free-riding, and reciprocal behaviors in WiFi sharing scenarios.

To formalize these evolutionary dynamics, the system adopts the replicator dynamics equation [5]:

$$\dot{x}_i = x_i (f_i(\mathbf{x}) - \phi(\mathbf{x})), \quad \phi(\mathbf{x}) = \sum_{j=1}^n x_j f_j(\mathbf{x}) \quad (1)$$

where $\dot{x}_i$ denotes the rate of change in the proportion of users adopting strategy i in the total population; x_i represents the proportion of users choosing the system's three core strategies, with x_1, x_2, x_3 corresponding to cooperative, free-riding and reciprocal users respectively; $f_i(\mathbf{x})$ quantifies the points earned by users selecting

strategy i via WiFi sharing; and $\phi(\mathbf{x})$ represents the population average return, calculated as the weighted average of all strategy returns, with weights corresponding to the proportion of users adopting each strategy.

Through the connection between points and credit levels, the system provides positive incentives for users who share WiFi and negative constraints for free-riders, forming a closed loop of “cooperation benefits and betrayal losses”.

To strictly quantify user payoffs and drive the Replicator Equation, our system replaces the generalized Prisoner’s Dilemma payoff matrix with a concrete 3×3 asymmetric payoff matrix. This matrix precisely defines the interactions among the three strategies: Cooperation, Free-riding, and Reciprocity.

TABLE I
THE SPECIFIC PAYOFF MATRIX IN THE SANDBOX SYSTEM

Player A \ Player B	Cooperation	Free-riding	Reciprocity
Cooperation	(+3, +3)	(−2, +5)	(+1, +1)
Free-riding	(+5, −2)	(0, 0)	(−2, +1)
Reciprocity	(+1, +1)	(+1, −2)	(+1, +1)

The second-price auction (Vickrey auction), a classic application of mechanism design theory, has a core rule: “the highest bidder wins and pays the second-highest bid amount”. This mechanism is suitable for the allocation of high-quality WiFi nodes during peak hours on campus, ensuring the efficiency and fairness of resource allocation.

III. SYSTEM DESIGN

To empirically validate the theoretical game models discussed above, we designed and implemented a comprehensive campus WiFi sharing simulation system. This system serves as a digital testbed that translates abstract economic incentives—such as cooperative surplus, free-rider exploitation, and reciprocal sanctioning—into programmable, functional modules. By doing so, it effectively maps theoretical Nash Equilibriums and Evolutionary Stable Strategies onto a measurable software environment where real-time data collection and behavioral analysis become feasible.

A. Architecture

The system adopts a modern Browser / Server decoupled architecture based on RESTful APIs, which guarantees high cohesion and low coupling. Specifically, the front-end is developed using HTML5, CSS3, and JavaScript, ensuring a lightweight and cross-platform user experience, while data visualization is powered by Chart.js for real-time rendering of evolutionary

graphs. The back-end is powered by FastAPI, a high-performance, asynchronous Python web framework. FastAPI naturally supports concurrent processing, which is highly suitable for handling high-frequency bidding requests during peak network congestion. For the data persistence layer, a lightweight relational database (SQLite) is utilized to ensure ACID properties for sensitive operations, particularly token transactions and reputation ledgers. The architecture closely follows the evolutionary game principles established in prior work [4], [5], embedding strategy adaptation as a first-class primitive.

B. Token and Reputation Mechanism

To prevent the tragedy of the commons, the system employs a dual-asset model: Tokens (Points) as the circulating currency and Reputation (Credit) as the long-term status indicator. Tokens are required to purchase bandwidth. To foster user retention, a continuous daily sign-in mechanism is introduced. The reward scales non-linearly with consecutive logins, creating a behavioral inertia that keeps users engaged in the game. Furthermore, reputation is calculated in real-time based on a user’s historical contribution and consumption. Let R denote the reputation score, S_{gb} be the total shared bandwidth, and C_{gb} be the total consumed bandwidth. The dynamic evaluation formula is defined as:

$$R = \min(100, \max(0, 80 + 2S_{gb} - C_{gb})) \quad (2)$$

This asymmetric formula generates a cooperative surplus by rewarding sharing twice as much as penalizing consumption. Based on R , users are categorized into four tiers: S-tier ($R \geq 90$), A-tier ($75 \leq R < 90$), B-tier ($60 \leq R < 75$), and C-tier ($R < 60$). Higher tier users earn visual badges and priority in the network, establishing a strong signaling mechanism. This design directly operationalizes the evolutionary game principle that cooperative behaviors should be rewarded while free-riding is penalized [4], [5].

C. Bandwidth Allocation Module

The bandwidth allocation module handles the core transactions between sharers and consumers through a closed-loop economic design. When a user publishes an idle WiFi node, they specify the bandwidth speed, reserved capacity, and access cost. To overcome the initial reluctance to share, the system immediately credits the publisher with a fixed token reward. During peak hours, when demand vastly exceeds supply, the system triggers a Generalized Second-Price (GSP) auction. Users submit sealed bids indicating their maximum willingness to pay, backed by a strict anti-overdraft validation on the

server. If a user outbids the market threshold, they win the bandwidth but are only charged the second-highest price. This ensures truthful bidding remains the dominant strategy, a property extensively studied in auction-based resource allocation [3], [6].

D. Multi-Agent Evolutionary Dashboard

The most prominent feature of the system is the Multi-Agent System (MAS) instantiated in the backend memory, which simulates the macroscopic emergence of strategies driven by the Replicator Dynamics equation. The backend engine initializes a population of independent virtual agents, each assigned a baseline strategy—Cooperation, Free-riding, or Reciprocity. The engine iteratively simulates game cycles where agents evaluate their payoffs against the population’s average. With a predefined probability factor, agents undergo strategy mutation. For instance, if the proportion of Free-riders excessively exceeds Cooperators, Cooperators will mutate into Free-riders out of exploitation aversion. Conversely, if a Free-rider’s reputation drops to a catastrophic level, the system forces them to revert to Cooperation. To visualize this, the frontend dashboard asynchronously polls the server, parses the real-time demographic distribution, and dynamically updates the charts, allowing researchers to observe the system stabilizing at an Evolutionary Stable Strategy (ESS). This instantiation directly realizes the replicator dynamics model [5].

IV. IMPLEMENTATION AND TESTING

A. Multi-Agent Evolution Engine

The system embeds an evolution engine that simulates 30 user agents, each characterized by one of three strategies: Cooperation, Free-riding, and Reciprocity. These agents interact in a simulated environment where their payoffs are determined by the payoff matrix defined in Table I. The engine dynamically updates the group’s strategy and reputation distribution in real-time according to the replicator dynamics equation, allowing the population to evolve toward equilibrium over successive iterations.

The frontend provides an evolution monitoring panel (as shown in Fig. 1), enabling intuitive observation of strategy proportion changes and reputation level distribution. Simulation verification shows that after 50 cycles, the system converges to an Evolutionary Stable Strategy (ESS): 70.00% Cooperators, 26.67% Reciprocators, and 0.03% Free-riders. Free-riders with low reputation are forced to switch to the cooperative strategy, effectively suppressing the “tragedy of the commons” and verifying the effectiveness of the mechanism design.

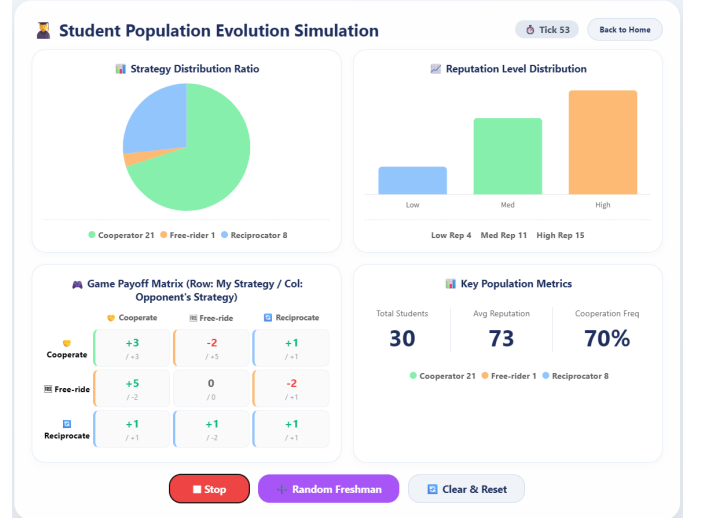


Fig. 1. Multi-agent strategy evolution monitoring interface.

B. Dual Mechanism Module

The system adopts a dual incentive system combining Points and Reputation: points serve as the circulating currency, while reputation determines user levels and access privileges. This dual-asset design creates a self-reinforcing loop where active sharing yields immediate token rewards and long-term reputation growth, while free-riding leads to reputation degradation. The community contribution list, shown in Fig. 2, displays each user’s real-time points balance, reputation score, and statistics of shared and consumed traffic.

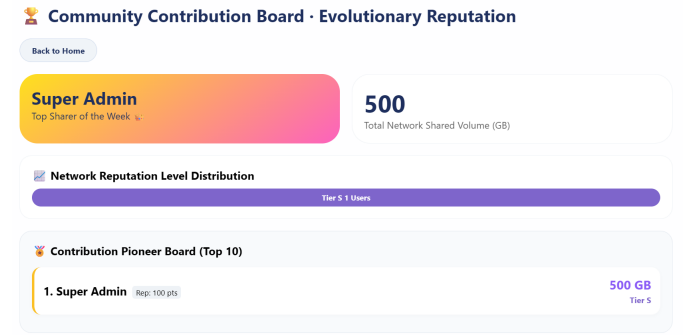


Fig. 2. The community contribution list.

Reputation levels are updated according to the asymmetric formula defined in Equation 2. Users are categorized into four tiers—S, A, B, and C—based on their reputation scores, with high-reputation users gaining priority access during peak hours and receiving visual badges. This design incentivizes users to maintain continuous sharing, forming a stable cooperative ecosystem where cooperation is both rewarded and recognized by peers.

C. WiFi Sharing and Peak-Hour Auction Module

Users can publish idle WiFi nodes by setting bandwidth and reserved capacity, and receive immediate point rewards upon publication (Fig. 3, left panel). For high-quality nodes during peak hours, the system employs the GSP auction mechanism: users submit bids, and the winner pays the second-highest price, encouraging truthful bidding. While auction-based allocation has been studied in heterogeneous networks [3], [6], our implementation extends it by adding reputation-based access control as an allocation criterion.

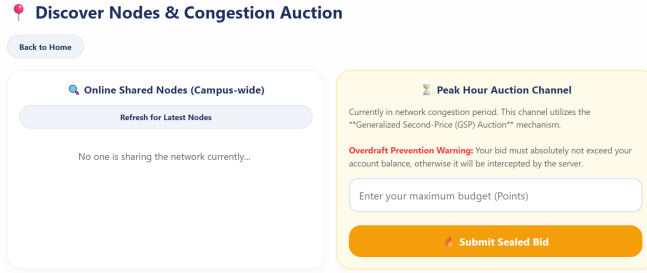


Fig. 3. WiFi sharing publication interface.

D. Points Redemption and Check-In Module

The system features a points redemption mall where users can exchange points earned from sharing for campus benefits such as milk tea coupons and printing vouchers as shown in Fig. 4 and Fig. 5. It also supports continuous check-in to obtain additional points, forming a closed loop of “Sharing–Profiting–Incentivizing”.

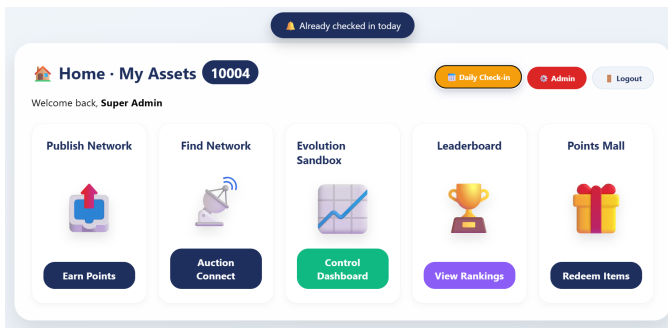


Fig. 4. Daily check-in.



Fig. 5. Points redemption interface.

Tests confirm that points deduction, redemption notifications, and check-in rewards function correctly, enhancing user engagement and system stability.

V. CONCLUSION

This paper designs and implements a campus WiFi sharing game theory sandbox system to address the low efficiency of campus WiFi resource sharing and the prevalence of free-riding behaviors. The test results demonstrate that the system, adopting a modern front-end and back-end separation architecture, successfully realizes core functionalities such as WiFi sharing, evolutionary game visualization, and Vickrey (second-price) auctioning. The integral and credit mechanism effectively guides users toward cooperative strategies, while the generalized second-price (GSP) auction mechanism optimizes the allocation of high-quality WiFi resources during peak hours.

In the future, the system will be optimized and expanded from three main engineering and theoretical dimensions. First, regarding the reputation algorithm, we plan to transition from the current real-time cumulative calculation to an Epoch-based Settlement mechanism. A Sliding Window Algorithm will be introduced to evaluate users' recent active behaviors, allowing historical reputation to gradually decay over time, thereby effectively forcing continuous contribution. Second, at the network physical layer, we aim to integrate actual Quality of Service (QoS) APIs to enforce bandwidth limits dynamically and extend the game model to realistic routing priorities. Third, at the system architecture level, we will replace the current HTTP polling mechanism with WebSocket full-duplex communication to achieve lower-latency and more efficient real-time synchronization of the multi-agent evolutionary dashboard.

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